

Optimizing Communication Strategies for COPD Management: Effectiveness of Educational Video and Pamphlet Interventions

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Introduction

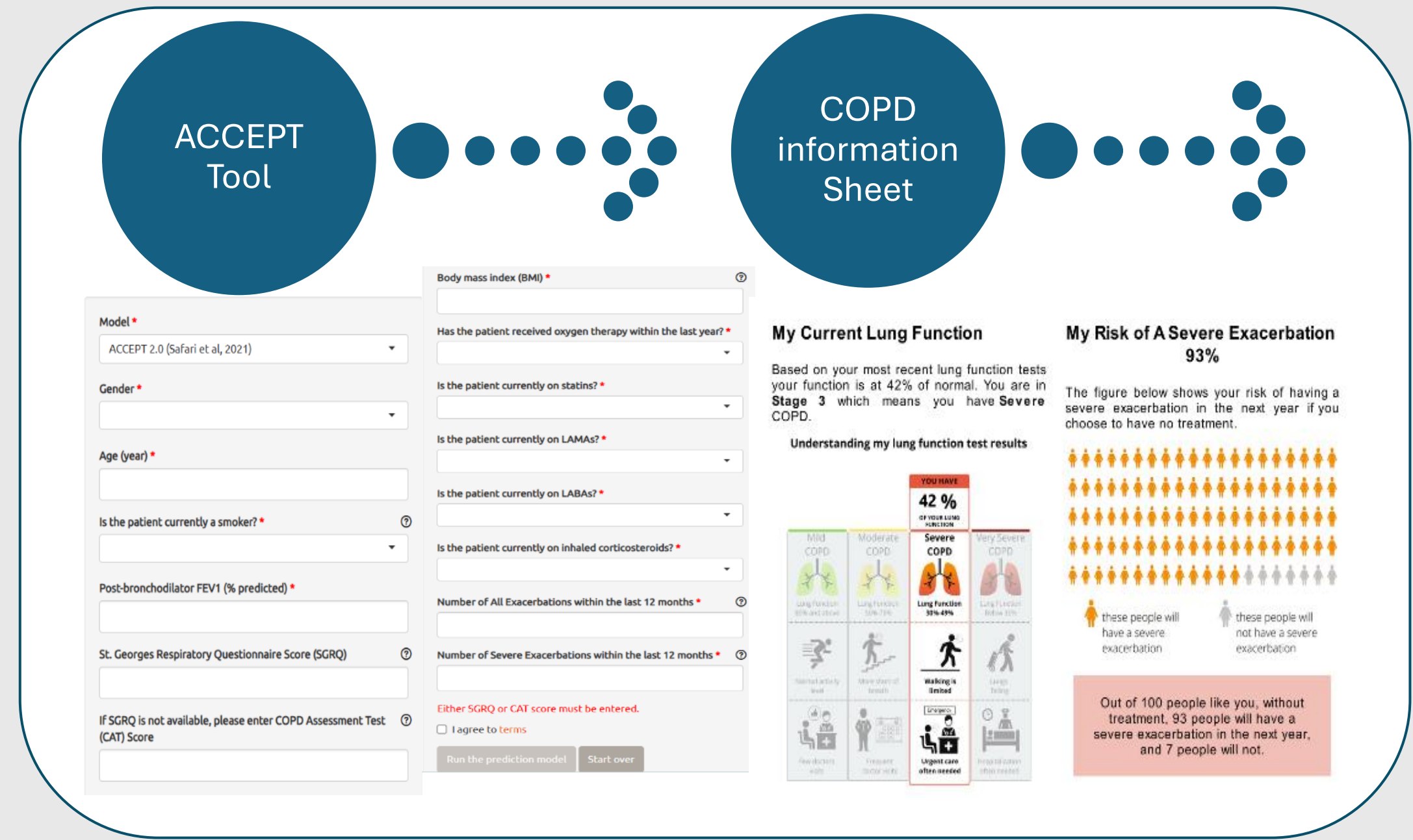
- Risk prediction tools are transforming care by supporting personalized treatment decisions.
- Yet, effectively communicating predictions from these tools to patients remains a challenge due to complex language and limited consultation time.

Chronic Obstructive Pulmonary Disease (COPD)

- Treatable, progressive lung disease.
- Affects over 2 million Canadians¹.
- Second leading cause of hospitalization in Canada².
- Exacerbations (flare-ups) in COPD are episodes of worsening of symptoms, leadings to substantial morbidity and mortality.

Acute COPD Exacerbation Prediction Tool (ACCEPT)³

Projects individualized rate and severity of COPD exacerbations.



This study evaluated different communication methods developed to improve general understanding of a risk prediction tool in COPD management.

Methods

Two communication materials

- A 3-minute video
- A pamphlet

Contents Covered

- The burden of COPD
- How a risk prediction model estimates risk of exacerbations

Participants

- Students and researchers in public health

Patient Partner Involvement

- Co-created the video and pamphlet
- Ensured materials are patient-centered and relevant for use as a support material in a clinic

Outcome

- Understanding of the precision medicine tool and preferred format

Results

14 Participants



71% Female



78% aged 24-64 years old

Pamphlet

93% believed the content was short, concise and straight to the point.

64% expressed that the visuals effectively depict a range of characters and roles.

86% found it simple, relevant and easy to understand.

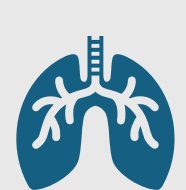
86% also found the length of the pamphlet about right.



29% White, 35% Asian



64% public health students



28% from the field of respiratory health

Video

64% believed style and presentation of the video content were well executed.

50% of viewers reported that the physician and the patient in the video were engaging and held their attention.

71% found it simple, relevant and east to understand.

36% found the duration of the video to be long.

After reviewing the material



All participants agreed that they became familiar with the benefits of precision medicine.



79% felt they received sufficient information about the prediction tool.



86% believed similar materials could support patient-provider conversations about precision medicine.

Suggestions for improvement

Video

- Better audio balance, more visual aids, and clearer speaker introductions.

Pamphlet

- More emphasis on the tool itself over background info.
- Suggestions on formatting (e.g., bold, color, bullet points) to enhance readability, and refining images to better support the text.

“For me, the video and the pamphlet are complementary. Together, they create a more complete way to convey the message. The pamphlet provides information that people can refer back to, while the video brings the patient testimony.”

Preferred Method of communication:

59% preferred pamphlets while 41% preferred video.

Conclusions

Tailored, patient-centered materials can enhance understanding across diverse audiences.

Future Steps

- Refine content based on feedback and usability findings.
- Implement materials in outpatient clinics (e.g., waiting room TVs and printed pamphlets).
- Make educational resources open access to broaden reach and impact.



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Collaboration for Outcomes Research and Evaluation
Faculty of Pharmaceutical Sciences

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2. Canadian Institute for Health Information . Hospital stays in Canada – series. 2024.
3. The Lancet Respiratory Medicine. The Acute COPD Exacerbation Prediction Tool (ACCEPT): a modelling study. 2020

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